Rohan Kumar

Final Year Undergraduate, IIT Kanpur
Major: Biological Sciences and Bioengineering
Minor: Industrial and Management Engineering

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Rohan Kumar
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Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech	Indian Institute of Technology Kanpur	2023-Present	**6.8**/10.0
Intermediate/+2	CBSE(XII)	Sarvodaya Bal Vidyalaya Rly Clny, New Delhi	2023	93.4%
Matriculation	CBSE(X)	GBSSS No.2 Rly Cly, New Delhi	2021	80.6%

Scholastic Achievements

- Cleared the **NISM-Series-VIII** certification examination conducted under **SEBI** regulations in **Equity Derivatives**.
- Secured **Top-10 at Strategy Storm, IIT Guwahati**, among **2,000+ participants**, through strategic analysis.

Work Experience

RamanIQ.ai *May'26 - Jul'26*
Software Developer Intern

- Built a **full-stack document platform** with **Next.js & FastAPI**, enabling structured document management.
- Designed a **modular service architecture** separating frontend, APIs, data processing, and persistence components.
- Modeled complex document hierarchies using **PostgreSQL & SQLAlchemy** with indexed and relational data structures.
- Developed reusable **REST APIs** powering document search, asset management, and seamless frontend-backend workflows.
- Automated **document ingestion workflows** for multi-format parsing, chunking, metadata extraction, and persistence.

Research Experience

IPO Price Dynamics in the Indian Market *May'26 - Aug'26*
Prof. Abhinava Triphati | Dept. of Management Sciences

- Studied **1,000+ Indian IPOs** through **underpricing, information-asymmetry, and price-discovery** frameworks.
- Reviewed **seminal IPO literature** and derived **empirical hypotheses** around demand, pricing, and investor behavior.
- Built an **IPO-level database** integrating price bands, **QIB/ NII/RII demand**, anchors, promoter stakes, and listing data.
- Automated **DRHP/RHP extraction** using **Python, OCR, NLP, PyMuPDF, and pdfplumber** to structure disclosures.
- Built **data-cleaning and feature-engineering workflows** for **underpricing, returns, and investor demand** analysis.

Position of Responsibility

Head, Udghosh'26 *May'26 - Present*
Annual Sport Fest — IIT Kanpur

- Spearheaded an 85+ member, **4-tier security team**, managing campus-wide deployment and **3 professional concert nights**.
- Coordinated with **Gymkhana, SIS, administration, and district authorities** for access control and crowd management.
- Oversaw security for **50+ events, social initiatives, and competitions**, ensuring seamless on-ground execution.
- Developed **security protocols** and **emergency procedures**, conducting drills to strengthen emergency preparedness.

Coordinator, Ultimate Frisbee Club *May'25 - May'26*
Games And Sports Council — IIT Kanpur

- Transformed the **Society into a Club**, managing an **INR 80k budget** and overseeing club operations and representation.
- Led the **Frisbee team as Captain**, organizing **3 intra-college tournaments** and driving **60%** participation growth.
- Initiated the Club's flagship inter-collegiate tournament, **IITK Invitationals**, securing **SBI as its inaugural sponsor**.
- Developed the Club's **official website** and built a **Frisbee simulation tool** for tactical analysis and team training.

Projects

GDP Prediction *May'26-Jul'26*
ECO723: Prof.Sanjeev Kumar

- Forecasted **India's quarterly GDP growth** using macroeconomic indicators and historical data.
- Engineered a **macroeconomic dataset** with 12+ indicators, using correlation analysis to reduce multicollinearity.
- Trained **Random Forest and XGBoost models** on 20+ years of macroeconomic data with lag features and tuning.
- Achieved **90%+ predictive accuracy** in stable periods, generating insights for **investment decisions and policy**.

Interest Rate Curve & Swap Risk Engine *May'26-Jun'26*
Self Project

- Engineered a modular **C++17 pricing engine** using **OOPs and polymorphism** for reusable financial computations.
- Designed reusable **instrument, curve, and interpolation interfaces** with modular components for scalable extension.
- Built a robust **CSV data pipeline** for market-data ingestion, maturity parsing, curve construction, and result generation.
- Developed an analytical **sensitivity engine** using chain-rule gradient propagation for efficient risk computation and testing.

Mess Automation System *Jan'26-Apr'26*
CS253: Prof.Indranil Saha

- Architected a scalable **tri-service system** using **React.js, Node.js/Express, and FastAPI** for real-time automation.
- Engineered **facial recognition** with **OpenCV LBPH**, automating training, recognition, and **PostgreSQL sync**.
- Automated **backend workflows** for dues, rebates, meal bookings, and feedback using **REST APIs and PostgreSQL**.
- Built interactive **analytics dashboards** for attendance, extra-buying, and revenue metrics using **backend data**.

Portfolio Management & Asset Pricing *Jun'25 - Jul'25*
Self Project

- Developed a portfolio optimization framework integrating **Fama -French and Black-Litterman models** to forecast returns.
- Automated **financial-metric extraction** via web scraping to build **fundamental health scores** for portfolio analysis.
- Applied **NLP** to news and social-media data to generate **sentiment signals** and trigger adaptive portfolio recalibration.

Market Insights *May'24-Jul'24*
Finance & Analytics Club, Summer Project

- Analyzed **stock fundamentals, technical indicators, and options strategies** to evaluate investments.
- Applied **fundamental and technical analysis** using valuation, chart patterns, and trading signals.

Skills

- **Languages:** C, C++, Java, Python, JavaScript, SQL
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras
- **Data & Tools:** PostgreSQL, MySQL, MongoDB, Git, GitHub

Extracurricular Activities

- Secured **Second Runner-Up at Udghosh'25**, leading the institute **Ultimate Frisbee team as captain**.
- Secured **Bronze in Ultimate Frisbee** during Inferno'26 (Inter-Hall Tournament), leading the **Hall-3 team as captain**.

Relevant Courses

** Ongoing Courses*

Foundation of Financial Risk	Security Analysis for Investments	Fundamentals of Data Science & ML	Accounting and Finance*
Investment Banking*	Applied Probability & Statistics	Software Development	Portfolio Management